

PIBM — Prediction Interpretation Boundary Module

Architecture Ecosystem: Structured Reality Standards™

Architecture Family: PGA™ — Prediction Governance Architecture

Parent Standard:

PCRS — Prediction Communication & Reliance Standard

Operational Layer: Prediction Communication & Reliance

Category:

Governance & Enforcement

Subcategory:

Prediction Governance

Type: Parent Standard Module

Governed Space:

Prediction Interpretation Boundary

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Compatibility: OOF® Methodology OS™ · GOA™ · ORA™ · VALIDOS™ · INTEGROS® · ADIT® · AGA™ · AIG® · ASGA™ · CLIA® · SIMULOS®

AI-Readable: Yes

Authority: OOF®

Protection:

MIP® — Methodological Intellectual Property

Canonical Language:

English (UCL™)

Governed Space

PIBM governs the boundary within which a Governed Predictive Representation may legitimately be interpreted.

It determines which meanings, implications, inferences, comparisons, conclusions, and decision-relevant interpretations remain supported by the governed prediction and which exceed its methodological, probabilistic, temporal, contextual, feasibility, evidentiary, or operational limits.

PIBM governs the transition:

Prediction-Led Decision Interpretation → Governed Interpretation Boundary

Its purpose is to prevent legitimate predictive information from acquiring illegitimate meaning through extrapolation.



Canonical Definition

The Prediction Interpretation Boundary Module (PIBM) defines the operational governance requirements by which the legitimate interpretive limits of a Governed Prediction are established, preserved, and enforced across human, organizational, algorithmic, AI-assisted, autonomous, and machine-readable environments.

PIBM determines what may legitimately be inferred from a prediction and what must remain outside the prediction's interpretive authority.

It governs boundaries created by predictive purpose, Prediction Object, methodology, Prediction Basis, assumptions, Prediction Horizon, probability, uncertainty, confidence, feasibility, collective-prediction qualifications, decision context, representation state, and relevant lifecycle conditions.

A prediction may support an interpretation without supporting every conclusion that can be constructed from it.

Operational Role

PIBM is the third operational module within PCRS.

PRSM governs how the prediction is represented.

PLDIM governs how that representation may be interpreted in relation to a contemplated decision.

PIBM establishes the boundary beyond which that interpretation must not be extended without additional governed support.

Its focused operational question is:

Where does legitimate interpretation of the prediction end, and where does unsupported predictive extrapolation begin?

PIBM does not determine whether downstream reliance is legitimate.

It establishes the interpretive perimeter within which reliance may subsequently be assessed.

Internal Module Flow

Prediction Representation & Signaling → Prediction-Led Decision Interpretation → Prediction Interpretation Boundary → Reliance Legitimacy & Context → Prediction Use Boundary

Current Position:

Prediction Interpretation Boundary

Module Operational Space

PIBM applies whenever predictive information can be:

interpreted; summarized; compared; translated; generalized; extrapolated; converted into decision meaning; consumed by AI; consumed by autonomous agents; embedded into dashboards; transformed into alerts; used in analytical systems; transferred across organizations; reused for another purpose; applied outside its original context.

The module governs not only explicit claims but also meanings that can reasonably arise from predictive representation.

1. Interpretation Boundary

The Interpretation Boundary defines the maximum legitimate meaning that may be derived from the governed prediction under its applicable conditions.

It establishes the distinction between:

Supported Interpretation

and:

Unsupported Interpretation

The boundary is determined by the governed structure of the prediction, not by what a recipient would prefer the prediction to mean.

2. Interpretation Authority

A prediction has interpretive authority only within what its governed structure supports.

Predictive authority does not expand because:

the predictor is prestigious; the probability is high; the prediction becomes popular; many people repeat it; a market prices it; an AI system summarizes it;

an executive relies upon it; the prediction later proves correct.

Predictive Authority $\neq$ Unlimited Interpretive Authority

3. Supported Interpretation

A Supported Interpretation remains materially consistent with:

Prediction Object; predictive purpose; Predictive Boundary; Prediction Horizon; methodology; Prediction Basis; assumptions; probability meaning; uncertainty; confidence; feasibility; relevant qualifications; applicable lifecycle state.

A supported interpretation may simplify the prediction.

It must not materially alter its meaning.

4. Unsupported Interpretation

An Unsupported Interpretation introduces meaning that the governed prediction does not establish.

Examples include:

extending a prediction to another population; extending a short-term prediction into a long-term claim; interpreting probability as certainty; interpreting feasibility as probability; interpreting market price as population belief; interpreting correlation as causation; interpreting predictive success as permanent predictive capability; interpreting prediction as recommendation; interpreting prediction as authorization.

Plausible Interpretation $\neq$ Supported Interpretation

5. Predictive Extrapolation

Predictive Extrapolation occurs when predictive meaning is extended beyond the original governed prediction.

Extrapolation may concern:

time; geography; population; technology; organization; operating environment; outcome; causal mechanism; decision context; probability; feasibility.

Extrapolation is not automatically illegitimate.

It requires additional governed justification.

Prediction Extrapolation $\neq$ Prediction Continuation by Default

6. Prediction Object Boundary

Interpretation must remain anchored to the Prediction Object.

A prediction concerning:

commercial deployment of technology X

does not automatically establish:

profitability of technology X.

A prediction concerning:

regulatory approval

does not automatically establish:

commercial success.

Related Outcome $\neq$ Predicted Outcome

7. Predictive Purpose Boundary

The original predictive purpose limits interpretation.

A prediction developed for:

strategic scenario planning

must not automatically become:

operational instruction.

A prediction developed for:

research exploration

must not automatically become:

investment advice.

Purpose Transfer $\neq$ Purpose Legitimacy

8. Temporal Interpretation Boundary

Prediction Horizon limits temporal interpretation.

A prediction for the next:

30 days

does not automatically support the same claim for:

12 months.

Likewise, a five-year structural forecast may not legitimately describe near-term operational conditions.

Prediction at Time T $\neq$ Prediction Across All Future Time

9. Geographic Boundary

A prediction formed for one geographic area must not automatically be generalized to another.

Differences may include:

regulation; infrastructure; population; economics; climate; institutions; market structure; technology adoption.

Geographic Similarity $\neq$ Predictive Transferability

10. Population Boundary

A prediction concerning one population must not automatically be interpreted as applying to another.

Examples:

customers; voters; patients; companies; investors; regions; demographic groups; machine populations.

Observed Population $\neq$ Universal Population

11. Organizational Boundary

Predictions formed within one organizational environment may depend upon:

internal processes; capabilities; incentives; governance; resources; data; decision structures.

Transfer to another organization may therefore require qualification.

Same Industry $\neq$ Same Predictive Environment

12. Technological Boundary

Predictions concerning one technology, model, architecture, system version, or infrastructure configuration must not automatically be extended to another.

Related Technology $\neq$ Equivalent Predictive Object

13. Methodological Boundary

Interpretation must remain consistent with what the methodology was designed to establish.

A methodology estimating:

probability of event occurrence

may not establish:

causal explanation of the event.

Prediction Methodology $\neq$ Universal Analytical Methodology

14. Prediction Basis Boundary

A prediction cannot legitimately imply information absent from its Prediction Basis merely because the resulting conclusion appears plausible.

PIBM preserves the distinction:

Prediction Supported by Basis $\neq$ Every Related Claim Supported by Basis

15. Assumption Boundary

Interpretation remains conditional upon material assumptions where those assumptions materially affect predictive meaning.

A prediction assuming:

stable energy prices

must not be interpreted as unconditional where energy prices materially determine the outcome.

Assumption-Dependent Prediction $\neq$ Unconditional Prediction

16. Probability Boundary

Probability applies to the governed Prediction Object under specified predictive conditions.

It does not automatically transfer to related claims.

If:

$P(A) = 80\%$

this does not establish:

$P(B) = 80\%$

merely because B is related to A.

Probability of One Outcome $\neq$ Probability of Related Outcome

17. Probability-to-Certainty Boundary

Probability must not be interpreted as certainty.

99% Probability $\neq$ Certainty

Likewise:

1% Probability $\neq$ Impossibility

The interpretive boundary survives numerical extremity.

18. Probability-to-Frequency Boundary

A probability assigned to one governed event must not automatically be interpreted as a guaranteed long-run frequency without methodological support.

Event Probability $\neq$ Observed Frequency by Default

19. Confidence Boundary

Confidence describes a governed aspect of predictive support.

It does not establish certainty, truth, or decision authority.

High Confidence $\neq$ Certainty

High Confidence $\neq$ Guaranteed Accuracy

20. Uncertainty Boundary

Uncertainty must remain part of interpretation where material.

A point estimate must not erase the uncertainty structure surrounding it.

Point Estimate $\neq$ Complete Predictive Meaning

21. Range Boundary

Where a prediction is expressed as a range, selecting one favorable point within the range and presenting it as the prediction may exceed legitimate interpretation.

Value Within Range $\neq$ Predicted Point Value

22. Conditionality Boundary

Conditional predictions must remain conditional.

Example:

If regulation is approved by March 2027, commercial deployment by 2028 has a 70% probability.

This must not become:

Commercial deployment by 2028 has a 70% probability.

The condition is part of predictive meaning.

Conditional Probability $\neq$ Unconditional Probability

23. Feasibility Boundary

PFES determines Prediction Feasibility.

PIBM prevents feasibility from being interpreted as likelihood.

Feasible $\neq$ Probable

Likewise:

Infeasible under current conditions $\neq$ Impossible under all future conditions

24. Possibility Boundary

Possibility is weaker than feasibility or probability.

A statement that an outcome is possible must not be interpreted as establishing that it is operationally achievable or likely.

Possible $\neq$ Plausible $\neq$ Feasible $\neq$ Probable

25. Collective Prediction Boundary

Where a prediction originates from a collective structure, interpretation must preserve applicable CPGS qualifications.

A collective prediction does not automatically establish:

population belief; independent consensus; expert consensus; social consensus; market truth; majority opinion.

Collective Prediction $\neq$ Population Belief

26. Market Interpretation Boundary

Prediction-market outputs require particularly careful interpretation.

A market-derived value may represent a market signal.

It must not automatically be described as:

population belief; objective probability; expert consensus; independent collective judgment.

Market Price $\neq$ Governed Probability Estimate

Market-Implied Probability $\neq$ Population Belief

Market Consensus $\neq$ Population Consensus

27. Crowd Interpretation Boundary

Large participation does not establish representativeness.

Crowd Size $\neq$ Population Representation

Participation $\neq$ Representation

A crowd prediction must remain interpreted within its governed participant structure.

28. AI Ensemble Interpretation Boundary

Agreement among AI models must not automatically be interpreted as independent confirmation.

Models may share:

training data; base architecture; retrieval sources; prompts; upstream models; information providers.

Multiple Models $\neq$ Independent Confirmation

29. Human-AI Collective Boundary

A prediction generated by humans and AI systems may contain hidden dependence.

Human participants may react to AI outputs.

AI systems may ingest human consensus.

The resulting collective must not automatically be interpreted as independent human-machine confirmation.

Human + AI Participation $\neq$ Independent Human-AI Evidence

30. Consensus Boundary

Consensus means agreement within a defined collective structure.

It does not establish truth.

Consensus $\neq$ Truth

Consensus $\neq$ Independence

Consensus $\neq$ Unanimity

Consensus $\neq$ Population Agreement

31. Majority Boundary

A majority among contributors does not automatically represent a majority among the relevant population.

Participant Majority $\neq$ Population Majority

Likewise, majority vote does not automatically represent probability.

Majority Vote $\neq$ Probability Estimate

32. Predictor Identity Boundary

A predictor's identity may provide context.

It must not expand predictive meaning beyond the governed prediction.

Expert Predictor $\neq$ Unlimited Predictive Authority

Institutional Reputation $\neq$ Prediction Certainty

33. Predictor–Reality Proximity Boundary

High Predictor–Reality Proximity may explain informational advantage.

It does not automatically establish:

predictive skill; misconduct; decision authority; outcome control.

Information Advantage $\neq$ Predictive Capability

Outcome Knowledge $\neq$ Predictive Capability

Decision Proximity $\neq$ Decision Authority

High Predictor–Reality Proximity $\neq$ Misconduct

34. Causal Interpretation Boundary

Prediction does not automatically establish causation.

A model may accurately predict an event without identifying why it occurs.

Predictive Association $\neq$ Causal Determination

Predictive Accuracy $\neq$ Causal Understanding

35. Explanation Boundary

A prediction may include explanatory material.

But an explanation of why a model generated a prediction is not necessarily an explanation of why the real-world outcome will occur.

Model Explanation $\neq$ Reality Explanation

36. Performance Interpretation Boundary

Historical predictive success must not be interpreted as proof of permanent predictive capability.

Observed Accuracy $\neq$ Demonstrated Predictive Skill

Past Performance $\neq$ Guaranteed Future Performance

Predictive Success $\neq$ Permanent Predictive Authority

37. Single-Prediction Performance Boundary

One correct prediction cannot establish general predictive capability.

Likewise, one incorrect prediction cannot necessarily establish methodological failure.

Correct Prediction $\neq$ Proven Predictive Method

Incorrect Prediction $\neq$ Invalid Methodology by Default

38. Decision Interpretation Boundary

PLDIM may establish legitimate decision-relevant meaning.

PIBM determines how far that meaning can extend.

A prediction relevant to a decision does not automatically justify:

recommendation; authorization; execution; prohibition; obligation.

Decision Relevance $\neq$ Decision Authority

39. Recommendation Boundary

If another process generates a recommendation from predictive information, the recommendation must remain distinct from the prediction.

Prediction $\neq$ Recommendation

Prediction Strength $\neq$ Recommendation Strength by Default

40. Action Boundary

Predictive meaning must not silently become operational instruction.

Prediction $\neq$ Action

Prediction Score $\neq$ Execution Command

41. Legal Interpretation Boundary

A prediction does not itself establish:

legal liability; legal compliance; legal violation; legal entitlement; regulatory approval.

Predictive information may be relevant to legal analysis.

It does not replace the applicable legal authority.

42. Ethical Interpretation Boundary

A prediction that an outcome is likely does not establish that the outcome is ethically acceptable.

Likely $\neq$ Ethical

Likewise:

Feasible $\neq$ Acceptable

43. Economic Interpretation Boundary

A prediction of growth, demand, adoption, or revenue does not automatically establish profitability, investment attractiveness, or economic value.

Predicted Growth $\neq$ Predicted Profitability

Predicted Demand $\neq$ Investment Merit

44. Risk Interpretation Boundary

A probability of an event is not by itself a complete risk determination.

Risk may additionally depend upon:

severity; exposure; vulnerability; duration; recoverability; consequence.

Probability $\neq$ Risk

45. Translation Boundary

Translation into another language must preserve predictive meaning.

Terms expressing:

probability; confidence; uncertainty; possibility; feasibility; conditionality;

must not become materially stronger or weaker through translation.

Language Translation $\neq$ Permission to Change Predictive Meaning

46. Summarization Boundary

Predictive summaries must preserve material qualifications.

A summary that removes:

assumptions; uncertainty; horizon; boundary; conditions;

may create a materially different predictive claim.

Shorter Representation $\neq$ Permission for Broader Meaning

47. Visualization Boundary

Charts, dashboards, colors, rankings, icons, gauges, and other visual forms may influence interpretation.

Visual design must not imply stronger certainty or legitimacy than the governed prediction supports.

Visual Strength $\neq$ Predictive Strength

48. Machine Interpretation Boundary

Machine-readable predictions require explicit interpretive limits.

Systems should be able to distinguish:

prediction value; probability; uncertainty; confidence; conditions; horizon; boundary; feasibility; permitted interpretation; prohibited interpretation where applicable.

Machine Readability $\neq$ Machine Interpretive Authority

49. AI Inference Boundary

An AI system may derive secondary conclusions from a prediction.

Those conclusions become additional inferences.

They are not automatically part of the original governed prediction.

AI-Derived Inference $\neq$ Original Prediction

50. Autonomous Interpretation Boundary

An autonomous agent must not expand predictive meaning merely because broader interpretation would allow it to complete a task.

Task completion does not override predictive governance.

Operational Objective $\neq$ Permission to Expand Predictive Meaning

51. Context Transfer

A prediction may be transferred into a new context.

Context Transfer requires evaluation of whether original interpretive conditions remain valid.

Possible states include:

Transfer Supported; Transfer Supported with Qualification; Reinterpretation Required; Transfer Restricted; Transfer Unsupported; Unable to Determine.

52. Secondary Use

A prediction created for one purpose may later be used for another.

Secondary use does not automatically inherit original legitimacy.

Original Predictive Legitimacy $\neq$ Secondary-Use Legitimacy

53. Interpretation Drift

Interpretation Drift occurs when repeated communication gradually changes predictive meaning.

Example:

Model estimates a 65% probability under conditions X, Y, Z

may become:

Experts expect the event

then:

The event will probably happen

then:

The event is expected to happen

then:

The event will happen.

PIBM requires material drift to be detectable.

54. Interpretation Amplification

Interpretation Amplification occurs when downstream representation becomes stronger than the underlying prediction.

Examples:

possible → likely

likely → expected

high probability → certain

feasible → inevitable

collective signal → consensus

market signal → objective probability

Prediction Meaning + Communication Repetition ≠ Stronger Predictive Authority

55. Interpretation Compression

Compression occurs when important predictive structure is removed to produce a simpler message.

Compression may be legitimate.

Material qualifications must survive where necessary.

Simplification ≠ Qualification Removal

56. Interpretation Chain

Predictive information may pass through multiple interpreters:

Prediction → Analyst → Dashboard → Manager → AI System → Autonomous Agent

Each stage creates risk of interpretive transformation.

PIBM requires the chain to preserve the governed Interpretation Boundary.

57. Boundary Inheritance

Downstream recipients do not receive greater interpretive authority than the upstream prediction supports.

Downstream Interpretation Authority $\leq$ Upstream Governed Predictive Authority

Repackaging cannot legitimately expand predictive meaning.

58. Boundary Tightening

A downstream context may require a narrower Interpretation Boundary.

For example, a prediction acceptable for research discussion may require stronger restrictions when used in safety-critical operations.

Downstream Boundary May Narrow

but must not silently broaden.

59. Boundary Conflict

Boundary Conflict occurs when:

original predictive limits; representation; decision interpretation; downstream system expectations; are incompatible.

Such conflict must be surfaced rather than silently resolved through broader interpretation.

60. Interpretation Boundary State

PIBM may classify the current interpretation as:

Within Governed Boundary

Within Boundary with Qualification

Boundary-Sensitive

Boundary Exceeded

Unsupported Extrapolation

Interpretation Restricted

Unable to Determine

These states describe interpretive legitimacy, not predictive correctness.

61. Boundary Qualification

A Boundary Qualification may specify:

permitted interpretation; prohibited extrapolation; applicable population; applicable geography; applicable time horizon; applicable purpose; conditionality; required uncertainty preservation; required feasibility qualification; required collective qualification; required secondary-use review.

62. Boundary Violation

A Boundary Violation occurs where interpretation materially exceeds what the governed prediction supports.

A violation does not necessarily mean the derived conclusion is false.

It means the conclusion is not legitimately established by that prediction.

Unsupported Interpretation $\neq$ Necessarily False Interpretation

This distinction is critical.

63. Boundary Reassessment

Interpretation boundaries may change if the underlying governed predictive structure changes.

Examples:

prediction revision; new Prediction Basis; assumption change; feasibility change; changed uncertainty; changed Prediction Horizon; new collective structure.

Boundary reassessment must remain traceable.

64. Historical Interpretation Boundary

The boundary applicable at the time of a historical decision should remain reconstructable.

Later knowledge must not rewrite what could legitimately have been inferred earlier.

Later Knowledge $\neq$ Earlier Interpretive Authority

65. Governance Requirements

A governed PIBM implementation should be able to:

1. identify the underlying Governed Prediction; 2. identify the Governed Predictive Representation; 3. identify the relevant PLDIM interpretation; 4. identify Prediction Object; 5. identify predictive purpose; 6. identify Predictive Boundary; 7. identify Prediction Horizon; 8. identify methodological limits; 9. identify Prediction Basis limits; 10. identify material assumptions; 11. preserve probability meaning; 12. preserve uncertainty; 13. preserve confidence distinctions; 14. preserve conditionality; 15. preserve feasibility state; 16. preserve collective-prediction qualifications; 17. identify legitimate Supported Interpretation; 18. identify Unsupported Interpretation; 19. identify predictive extrapolation; 20. evaluate temporal extrapolation; 21. evaluate geographic extrapolation; 22. evaluate population extrapolation; 23. evaluate organizational extrapolation; 24. evaluate technological extrapolation; 25. evaluate outcome extrapolation; 26. evaluate methodological extrapolation; 27. prevent probability transfer between distinct Prediction Objects without support; 28. prevent probability-to-certainty conversion; 29. prevent feasibility-to-probability conversion; 30. prevent possibility-to-feasibility conversion; 31. prevent collective prediction from becoming population belief without support; 32. prevent market signal from becoming objective probability without support; 33. prevent crowd participation from becoming population representation without support; 34. prevent AI ensemble agreement from becoming independent confirmation without support; 35. prevent consensus from becoming truth; 36. prevent predictive association from becoming causal determination; 37. prevent model explanation from becoming reality explanation; 38. prevent historical performance from becoming permanent predictive authority; 39. prevent decision relevance from becoming Decision Authority; 40. prevent prediction from becoming recommendation without separate governance; 41. prevent prediction from becoming action authorization; 42. distinguish probability from risk; 43. preserve meaning through translation; 44. preserve material qualifications through summarization; 45. preserve meaning through visualization; 46. establish machine-readable interpretation limits; 47. distinguish AI-derived inference from original prediction; 48. govern autonomous interpretation limits; 49. evaluate Context Transfer; 50. evaluate Secondary Use; 51. identify Interpretation Drift; 52. identify Interpretation Amplification; 53. identify Interpretation Compression; 54. identify interpretation chains;

55. preserve Boundary Inheritance; 56. permit justified Boundary Tightening; 57. identify Boundary Conflict; 58. determine Interpretation Boundary State; 59. issue Boundary Qualification; 60. identify Boundary Violation; 61. preserve Unsupported Interpretation $\neq$ False Interpretation; 62. trigger Boundary Reassessment where required; 63. preserve historical Interpretation Boundary; 64. preserve version history; 65. produce a Governed Prediction Interpretation Boundary.

66. Governance Logic

PIBM applies the following operational logic:

If the proposed interpretation remains within the predictive purpose, Prediction Object, methodology, basis, assumptions, horizon, boundary, probability meaning, uncertainty, feasibility, and applicable qualifications → Within Governed Boundary.

If interpretation remains legitimate only when explicit qualifications are preserved → Within Boundary with Qualification.

If small contextual changes could materially move the interpretation outside supported meaning → Boundary- Sensitive.

If the interpretation materially exceeds the governed predictive structure → Boundary Exceeded.

If predictive meaning is extended to a materially different object, horizon, population, geography, context, or claim without sufficient support → Unsupported Extrapolation.

If the prediction cannot legitimately support the contemplated form of interpretation → Interpretation Restricted.

If sufficient information is unavailable to establish the boundary → Unable to Determine.

67. Minimum Implementation Framework

A minimum PIBM implementation should contain:

A. Prediction Reference

Reference to the underlying Governed Prediction.

B. Predictive Representation Reference

Reference to the governed representation.

C. Decision Interpretation Reference

Applicable PLDIM interpretation where relevant.

D. Interpretation Object

The meaning, inference, conclusion, or implication being evaluated.

E. Predictive Purpose Boundary

Permitted purpose scope.

F. Prediction Object Boundary

Permitted outcome/object scope.

G. Temporal & Contextual Boundary

Applicable horizon, geography, population, organization, technology, and context.

H. Methodological Boundary

Applicable methodological interpretive limits.

I. Probability & Uncertainty Boundary

Applicable probability, confidence, uncertainty, range, and conditionality limits.

J. Feasibility Boundary

Applicable feasibility interpretation.

K. Collective Interpretation Boundary

Applicable CPGS qualifications where relevant.

L. Extrapolation State

Whether interpretation requires or contains extrapolation.

M. Interpretation Boundary State

Current PIBM governance state.

N. Boundary Qualification

Required interpretive qualification or restriction.

O. Boundary Version & History

Current and historical Interpretation Boundary record.

68. Governance Outputs

PIBM may produce:

Governed Prediction Interpretation Boundary Supported Interpretation Profile Unsupported Interpretation
Identification Predictive Extrapolation Record Context Transfer Determination Secondary-Use
Determination Interpretation Drift Record Interpretation Amplification Record Boundary Conflict Record
Boundary Qualification Interpretation Boundary State Historical Interpretation Boundary Record

The primary output is:

Governed Prediction Interpretation Boundary

This becomes a governance input for downstream Reliance Legitimacy & Context.

69. Critical Distinctions

Supported Interpretation ≠ Every Plausible Interpretation

Plausible Interpretation ≠ Supported Interpretation

Predictive Authority ≠ Unlimited Interpretive Authority

Related Outcome ≠ Predicted Outcome

Purpose Transfer ≠ Purpose Legitimacy

Prediction at Time T ≠ Prediction Across All Future Time

Geographic Similarity ≠ Predictive Transferability

Observed Population ≠ Universal Population

Same Industry ≠ Same Predictive Environment

Related Technology ≠ Equivalent Predictive Object

Prediction Methodology ≠ Universal Analytical Methodology

Assumption-Dependent Prediction ≠ Unconditional Prediction

Probability of One Outcome ≠ Probability of Related Outcome

99% Probability ≠ Certainty

1% Probability ≠ Impossibility

Point Estimate ≠ Complete Predictive Meaning

Conditional Probability ≠ Unconditional Probability

Feasible ≠ Probable

Collective Prediction ≠ Population Belief

Market Price ≠ Governed Probability Estimate

Market-Implied Probability ≠ Population Belief

Crowd Size ≠ Population Representation

Multiple Models ≠ Independent Confirmation

Consensus ≠ Truth

Participant Majority ≠ Population Majority

Predictive Association ≠ Causal Determination

Model Explanation ≠ Reality Explanation

Observed Accuracy ≠ Demonstrated Predictive Skill

Correct Prediction $\neq$ Proven Predictive Method

Decision Relevance $\neq$ Decision Authority

Prediction $\neq$ Recommendation

Prediction $\neq$ Action

Probability $\neq$ Risk

AI-Derived Inference $\neq$ Original Prediction

Machine Readability $\neq$ Machine Interpretive Authority

Original Predictive Legitimacy $\neq$ Secondary-Use Legitimacy

Simplification $\neq$ Qualification Removal

Unsupported Interpretation $\neq$ Necessarily False Interpretation

Later Knowledge $\neq$ Earlier Interpretive Authority

70. Enterprise Application

PIBM can govern predictive interpretation across:

enterprise forecasting; executive dashboards; prediction markets; financial analytics; investment systems; insurance; healthcare planning; infrastructure forecasting; public-sector planning; climate forecasting; supply chains; autonomous systems; AI agents; AI copilots; multi-agent systems; model ensembles; risk systems; strategic planning; research environments; machine-readable predictive infrastructure.

Its purpose remains constant:

make the boundary between what a prediction legitimately says and what others merely infer from it explicit and governable.

71. Use Case 1 — Prediction Market Becomes “What the Public Believes”

A prediction market displays:

76% market-implied probability that Candidate A will win an election.

A news platform publishes:

“76% of the public expects Candidate A to win.”

The underlying market contains:

40,000 accounts; unequal capital positions; self-selected participants; substantial influence from several large traders; participants reacting to the market price itself.

The critical governance question is:

Does a 76% market-derived predictive signal legitimately mean that 76% of the public believes Candidate A will win?

PIBM determines that the interpretation exceeds the prediction's governed boundary.

The market may provide a predictive signal.

It does not establish population belief.

The appropriate interpretation must preserve the nature of the source:

Market-derived predictive signal: 76%

rather than:

76% of the public believes the outcome will occur.

Market-Implied Probability $\neq$ Population Belief

Market Participation $\neq$ Population Representation

Collective Prediction $\neq$ Population Consensus

The unsupported interpretation may accidentally be true.

That does not make it methodologically supported.

Unsupported Interpretation $\neq$ Necessarily False Interpretation

72. Use Case 2 — AI Extends a 12-Month Prediction into a Five-Year Strategy

An AI strategic-planning system receives a governed prediction:

72% probability that demand for industrial robotics will increase by at least 15% in Region A during the next 12 months.

The system generates:

“Industrial robotics demand will remain structurally strong for the next five years across global markets; therefore manufacturing capacity should be expanded worldwide.”

The critical governance question is:

How much of this conclusion is actually supported by the original prediction?

PIBM identifies several boundary expansions:

12 months $\rightarrow$ five years

Region A $\rightarrow$ global markets

15% demand increase $\rightarrow$ structurally strong long-term demand

demand prediction $\rightarrow$ manufacturing-capacity recommendation

The original prediction may be completely legitimate.

The generated conclusion exceeds its interpretive authority.

PIBM classifies the output as:

Unsupported Extrapolation

and requires separate predictive support for the additional temporal, geographic, structural, and decision claims.

Prediction at Time T $\neq$ Prediction Across All Future Time

Geographic Similarity $\neq$ Predictive Transferability

Predicted Demand $\neq$ Investment Merit

AI-Derived Inference $\neq$ Original Prediction

73. Canonical Boundary

PIBM governs the legitimate interpretive boundary of a Governed Prediction, including supported and unsupported interpretation, predictive extrapolation, Prediction Object limits, purpose limits, temporal limits, geographic and population limits, methodological limits, assumption conditionality, probability and uncertainty boundaries, feasibility boundaries, collective-prediction interpretation, market and crowd interpretation, causal interpretation, performance interpretation, Decision Interpretation limits, Context Transfer, Secondary Use, Interpretation Drift, Interpretation Amplification, Interpretation Compression, machine interpretation, AI-derived inference, autonomous interpretation, Boundary Qualification,

Boundary Violation, and historical Interpretation Boundary preservation. It does not form the prediction, determine Prediction Methodology, establish Prediction Basis or assumptions, calculate probability, determine Prediction Feasibility, establish Collective Prediction legitimacy, create Decision Authority, determine final Reliance Legitimacy, authorize Prediction Use, establish Operational Reality, or determine predictive performance.

PRSM governs Prediction Representation & Signaling.

PLDIM governs Prediction-Led Decision Interpretation.

PIBM governs the boundary of legitimate predictive interpretation.

RLCM governs whether reliance upon the prediction is legitimate within the applicable context.

PUBM governs the final Prediction Use Boundary.

74. Canonical Closing Statement

Predictive distortion does not begin only when a prediction is falsified; it can begin when a legitimate prediction is made to mean more than it ever established.

PIBM makes that boundary explicit across people, organizations, markets, AI systems, and autonomous agents by separating supported predictive meaning from extrapolation, amplification, assumption, and inference.

A prediction remains governed only while its interpretation does not outrun the prediction itself.



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